ABSTRACT OF THE DISCLOSURE

A system and method for high-radix network communication and real-time immutable archival. In a preferred embodiment, the system operates as an end-to-end natively optical high-radix network wherein information is natively generated, processed by optically-native central processing units, routed via pixel-based optical interconnects, and transmitted as high-radix multi-spectral state frames under dynamically rotating forward-secret parametric maps, destroying predictable cryptanalytic footprints across the network. In a transitional embodiment, the system retrofits legacy serial-binary devices via software/hardware translation codecs for backwards compatibility over legacy communication channels. Furthermore, the extreme data density of native high-radix frames enables real-time physical dumping directly to immutable optical media, bypassing serial-binary conversion bottlenecks.